Research Analysis Report

Research Topic: System Development Intelligence

Total Papers: 5

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Executive Summary

This report provides a technical overview of 'System Development Intelligence' including methodologies, research gaps and experiment planning.

Literature Review

Introduction: This literature review summarizes findings from 5 research papers related to the selected topic. It highlights research trends, methodologies, key contributions, future directions, and overall observations.

Paper Summaries

• {'title': 'Development and validation of an autonomous artificial intelligence agent for clinical decision-making in oncology', 'year': 2025, 'summary': 'Clinical decision-making in oncology is complex, requiring the integration of multimodal data and multidomain expertise. We developed and evaluated an autonomous clinical artificial intelligence (AI) agent leveraging GPT-4 with multimodal precision oncology tools to support personalized clinical decision-making. The system incorporates vision transformers for detecting microsatellite instability and KRAS and BRAF mutations from histopathology slides, MedSAM for radiological image segmentation and web-based search tools such as OncoKB, PubMed and Google. Evaluated on 20 realistic multimodal patient cases, the AI agent autonomously used appropriate tools with 87.5% accuracy, reached correct clinical conclusions in 91.0% of cases and accurately cited relevant oncology guidelines 75.5% of the time. Compared to GPT-4 alone, the integrated AI agent drastically improved decision-making accuracy from 30.3% to 87.2%. These findings demonstrate that integrating language models with precision oncology and search tools substantially enhances clinical accuracy, establishing a robust foundation for deploying AI-driven personalized oncology support systems. Ferber et al. present an autonomous artificial intelligence agent system for deployment of specialized medical oncology computational tools, validating their system across various clinical scenarios representative of typical patient care workflows.'}

• {'title': 'Artificial intelligence in sustainable development research', 'year': 2025, 'summary': 'Artificial intelligence (AI) holds significant potential to advance Sustainable Development Goals by enabling data-driven insights and optimizations. In this analysis, we review 792 articles that explore AI applications in Sustainable Development Goal-related research. The literature is organized along two dimensions: (1) the disciplinary spectrum, from natural sciences to the humanities, and (2) the focus, distinguishing economic from socioecological content. Deep learning and supervised machine learning were the most prominently applied algorithms for forecasting and system optimization. However, we identify a critical gap: only a few studies combine advanced AI applications with deep sustainability expertise. Sustainability needs to strike a balance between contextualization and generalizability to provide tangible knowledge that will lead to responsible change. AI must play a central role in this process. While expectations for AI’s transformative role in sustainable development are high, its full potential remains to be realized. The use of artificial intelligence (AI) in sustainability-related scholarly work, such as Sustainable Development Goal research, is growing. An analysis now finds that few studies actually use AI to address normative or transformative dimensions of sustainability science, limiting the potential of relevant AI applications.'}

• {'title': 'AI Governance in the System Development Life Cycle: Insights on Responsible Machine Learning Engineering', 'year': 2022, 'summary': 'In this study we explore the incorporation of artificial intelligence (AI) governance to system development life cycle (SDLC) models. We conducted expert interviews among AI and SDLC professionals and analyzed the interview data using qualitative coding and clustering to extract AI governance concepts. Subsequently, we mapped these concepts onto three stages in the machine learning (ML) system development process: (1) design, (2) development, and (3) operation. We discovered 20 governance concepts, some of which are relevant to more than one of the three stages. Our analysis highlights AI governance as a complex process that involves multiple activities and stakeholders. As development projects are unique, the governance requirements and processes also vary. This study is a step towards understanding how AI governance is conceptually connected to ML systems' management processes through the project life cycle. CCS CONCEPTS • Software and its engineering $^{\\rightarrow}$ Software creation and management.'}

• {'title': 'Meaningful human control: actionable properties for AI system development', 'year': 2021, 'summary': 'How can humans remain in control of artificial intelligence (AI)-based systems designed to perform tasks autonomously? Such systems are increasingly ubiquitous, creating benefits - but also undesirable situations where moral responsibility for their actions cannot be properly attributed to any particular person or group. The concept of meaningful human control has been proposed to address responsibility gaps and mitigate them by establishing conditions that enable a proper attribution of responsibility for humans; however, clear requirements for researchers, designers, and engineers are yet inexistent, making the development of AI-based systems that remain under meaningful human control challenging. In this paper, we address the gap between philosophical theory and engineering practice by identifying, through an iterative process of abductive thinking, four actionable properties for AI-based systems under meaningful human control, which we discuss making use of two applications scenarios: automated vehicles and AI-based hiring. First, a system in which humans and AI algorithms interact should have an explicitly defined domain of morally loaded situations within which the system ought to operate. Second, humans and AI agents within the system should have appropriate and mutually compatible representations. Third, responsibility attributed to a human should be commensurate with that human's ability and authority to'}

• {'title': 'Development of a real-time muck analysis system for assistant intelligence TBM tunnelling', 'year': 2021, 'summary': 'Abstract Intelligent tunnelling has become an important direction for the development of TBM technology recently. As a result of the interaction between rock mass and TBM cutterhead, mucks are very important for predicting rock mass conditions and evaluating rock breaking efficiency. A real-time muck analysis system for assistant intelligence TBM tunnelling is proposed in this paper. Machine vision was applied to take the muck images continuously in the high-speed conveyor belt. The image segmentation and feature extraction of the mucks are conducted by using a deep learning algorithm. The proposed system also measured the mass and volume flow of the muck by installing a belt scale and a scanner to monitor the stability of the rock mass on the tunnel face. After the system was completed, it was installed on an indoor simulation experimental platform. A series of experiments were conducted to verify the design functions and measurement accuracy. Additionally, the system was applied to a TBM tunnelling project. The application results showed that the proposed system reached its design requirements and functions, and can provide muck data support for further assistant intelligent TBM tunnelling.'}

Research Trends

• accuracy

• actionable

• address

• agent

- ai-based
- ai-driven
- application
- applied
- appropriate
- artificial
- assistant
- attributed
- autonomous
- belt
- clinical
- clustering
- concept
- conceptually
- conducted
- control
- cycle
- data-driven
- decision-making
- deep
- design
- development
- dimension
- function
- goal-related
- governance
- high-speed
- human
- image
- intelligence
- interview
- knowledge
- learning
- life
- making
- management
- mass
- meaningful

• muck
• mucks
• multimodal
• oncology
• optimization
• personalized
• project
• real-time
• remain
• responsibility
• rock
• sdlc
• situation
• software
• stage
• sustainability
• sustainability-related
• sustainable
• three
• tools
• transformative
• transformer
• tunnelling
• vision
• web-based

Research Gaps

• While expectations for AI’s transformative role in sustainable development are high, its full potential remains to be realized.

Future Scope

• While expectations for AI’s transformative role in sustainable development are high, its full potential remains to be realized.

Conclusion: Overall, the reviewed studies demonstrate significant progress in this research area while highlighting several opportunities for future work and further experimentation.

Methodology Comparison

Total Papers: 5

Comparison

• {'title': 'Development and validation of an autonomous artificial intelligence agent for clinical decision-making in oncology', 'research_area': 'Retrieval-Augmented Generation', 'research_problem': 'Clinical decision-making in oncology is complex, requiring the integration of multimodal data and multidomain expertise.', 'methodology': ['Experimental Evaluation', 'Retrieval-Augmented Generation', 'Transformer'], 'keywords': ['agent', 'autonomous', 'decision-making', 'transformer', 'clinical', 'oncology', 'vision', 'tools', 'multimodal', 'ai-driven', 'web-based', 'accuracy', 'artificial', 'intelligence', 'personalized'], 'year': 2025, 'citation_count': 171, 'quality_score': 90, 'quality_classification': 'Excellent'}

• {'title': 'Artificial intelligence in sustainable development research', 'research_area': 'Machine Learning', 'research_problem': 'However, we identify a critical gap: only a few studies combine advanced AI applications with deep sustainability expertise.', 'methodology': ['Algorithm', 'Machine Learning', 'Deep Learning'], 'keywords': ['knowledge', 'development', 'sustainable', 'data-driven', 'goal-related', 'sustainability-related', 'application', 'sustainability', 'artificial', 'intelligence', 'optimization', 'transformative', 'deep', 'dimension', 'learning'], 'year': 2025, 'citation_count': 71, 'quality_score': 80, 'quality_classification': 'Very Good'}

• {'title': 'AI Governance in the System Development Life Cycle: Insights on Responsible Machine Learning Engineering', 'research_area': 'Machine Learning', 'research_problem': 'In this study we explore the incorporation of artificial intelligence (AI) governance to system development life cycle (SDLC) models.', 'methodology': ['Machine Learning'], 'keywords': ['governance', 'development', 'concept', 'management', 'cycle', 'interview', 'life', 'project', 'sdlc', 'software', 'stage', 'three', 'artificial', 'clustering', 'conceptually'], 'year': 2022, 'citation_count': 42, 'quality_score': 70, 'quality_classification': 'Very Good'}

• {'title': 'Meaningful human control: actionable properties for AI system development', 'research_area': 'General Artificial Intelligence', 'research_problem': 'How can humans remain in control of artificial intelligence (AI)-based systems designed to perform tasks autonomously?', 'methodology': ['Algorithm'], 'keywords': ['human', 'ai-based', 'agent', 'responsibility', 'control', 'meaningful', 'attributed', 'address', 'making', 'remain', 'situation', 'actionable', 'application', 'appropriate', 'artificial'], 'year': 2021, 'citation_count': 101, 'quality_score': 70, 'quality_classification': 'Very Good'}

• {'title': 'Development of a real-time muck analysis system for assistant intelligence TBM tunnelling', 'research_area': 'General Artificial Intelligence', 'research_problem': 'Abstract Intelligent tunnelling has become an important direction for the development of TBM technology recently.', 'methodology': ['Algorithm', 'Experimental Evaluation', 'Deep Learning'], 'keywords': ['vision', 'tunnelling', 'high-speed', 'mass', 'muck', 'rock', 'real-time', 'applied', 'assistant', 'belt', 'conducted', 'design', 'function', 'image', 'mucks'], 'year': 2021, 'citation_count': 69, 'quality_score': 60, 'quality_classification': 'Good'}

Highest Cited Paper

title: Development and validation of an autonomous artificial intelligence agent for clinical decision-making in oncology

authors: ['Dyke Ferber', 'O. E. El Nahhas', 'G. Wölflein', 'I. Wiest', 'J. Clusmann', 'M. Leßmann', 'S. Foersch', 'Jacqueline Lammert', 'Maximilian Tschochohei', 'Dirk Jaeger', 'M. Salto-Tellez', 'N. Schultz', 'D. Truhn', 'J. Kather']

year: 2025

citation_count: 171

summary: Clinical decision-making in oncology is complex, requiring the integration of multimodal data and multidomain expertise. We developed and evaluated an autonomous clinical artificial intelligence (AI) agent leveraging GPT-4 with multimodal precision oncology tools to support personalized clinical decision-making. The system incorporates vision transformers for detecting microsatellite instability and KRAS and BRAF mutations from histopathology slides, MedSAM for radiological image segmentation and web-based search tools such as OncoKB, PubMed and Google. Evaluated on 20 realistic multimodal patient cases, the AI agent autonomously used appropriate tools with 87.5% accuracy, reached correct clinical conclusions in 91.0% of cases and accurately cited relevant oncology guidelines 75.5% of the time. Compared to GPT-4 alone, the integrated AI agent drastically improved decision-making accuracy from 30.3% to 87.2%. These findings demonstrate that integrating language models with precision oncology and search tools substantially enhances clinical accuracy, establishing a robust foundation for deploying AI-driven personalized oncology support systems. Ferber et al. present an autonomous artificial intelligence agent system for deployment of specialized medical oncology computational tools, validating their system across various clinical scenarios representative of typical patient care workflows.

research_problem: Clinical decision-making in oncology is complex, requiring the integration of multimodal data and multidomain expertise.

methodology: ['Experimental Evaluation', 'Retrieval-Augmented Generation', 'Transformer']

key_contributions: ['We developed and evaluated an autonomous clinical artificial intelligence (AI) agent leveraging GPT-4 with multimodal precision oncology tools to support personalized clinical decision-making.', 'These findings demonstrate that integrating language models with precision oncology and search tools substantially enhances clinical accuracy, establishing a robust foundation for deploying AI-driven personalized oncology support systems.']

future_work: []

keywords: ['agent', 'autonomous', 'decision-making', 'transformer', 'clinical', 'oncology', 'vision', 'tools', 'multimodal', 'ai-driven', 'web-based', 'accuracy', 'artificial', 'intelligence', 'personalized']

research_area: Retrieval-Augmented Generation

paper_score: 90

paper_quality: Excellent

Latest Paper

title: Development and validation of an autonomous artificial intelligence agent for clinical decision-making in oncology

authors: ['Dyke Ferber', 'O. E. El Nahhas', 'G. Wölflein', 'I. Wiest', 'J. Clusmann', 'M. Leßmann', 'S. Foersch', 'Jacqueline Lammert', 'Maximilian Tschochohei', 'Dirk Jaeger', 'M. Salto-Tellez', 'N. Schultz', 'D. Truhn', 'J. Kather']

year: 2025

citation_count: 171

summary: Clinical decision-making in oncology is complex, requiring the integration of multimodal data and multidomain expertise. We developed and evaluated an autonomous clinical artificial intelligence (AI) agent leveraging GPT-4 with multimodal precision oncology tools to support personalized clinical decision-making. The system incorporates vision transformers for detecting microsatellite instability and KRAS and BRAF mutations from histopathology slides, MedSAM for radiological image segmentation and web-based search tools such as OncoKB, PubMed and Google. Evaluated on 20 realistic multimodal patient cases, the AI agent autonomously used appropriate tools with 87.5% accuracy, reached correct clinical conclusions in 91.0% of cases and accurately cited relevant oncology guidelines 75.5% of the time. Compared to GPT-4 alone, the integrated AI agent drastically improved decision-making accuracy from 30.3% to 87.2%. These findings demonstrate that integrating language models with precision oncology and search tools substantially enhances clinical accuracy, establishing a robust foundation for deploying AI-driven personalized oncology support systems. Ferber et al. present an autonomous artificial intelligence agent system for deployment of specialized medical oncology computational tools, validating their system across various clinical scenarios representative of typical patient care workflows.

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methodology: ['Experimental Evaluation', 'Retrieval-Augmented Generation', 'Transformer']

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future_work: []

keywords: ['agent', 'autonomous', 'decision-making', 'transformer', 'clinical', 'oncology', 'vision', 'tools', 'multimodal', 'ai-driven', 'web-based', 'accuracy', 'artificial', 'intelligence', 'personalized']

research_area: Retrieval-Augmented Generation

paper_score: 90

paper_quality: Excellent

Common Methodologies

• Algorithm

• Deep Learning

• Experimental Evaluation

• Machine Learning

• Retrieval-Augmented Generation

• Transformer

Research Areas

• General Artificial Intelligence

• Machine Learning

• Retrieval-Augmented Generation

Common Keywords

• accuracy

• actionable

• address

• agent

• ai-based

• ai-driven

• application

• applied

• appropriate

• artificial

• assistant

• attributed

- autonomous
- belt
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- remain
- responsibility
- rock
- sdlc
- situation
- software
- stage
- sustainability
- sustainability-related
- sustainable
- three
- tools
- transformative
- transformer
- tunnelling
- vision
- web-based

Methodology Frequency

Algorithm: 3

Experimental Evaluation: 2

Machine Learning: 2

Deep Learning: 2

Retrieval-Augmented Generation: 1

Transformer: 1

Most Common Methodology: Algorithm

Quality Distribution

Excellent: 1

Very Good: 3

Good: 1

Publication Years

2021: 2

2022: 1

2025: 2

Citation Statistics

highest: 171

lowest: 42

average: 90.8

total: 454

Comparison Summary

total_papers: 5

most_common_methodology: Algorithm

research_areas: ['General Artificial Intelligence', 'Machine Learning', 'Retrieval-Augmented Generation']

quality_distribution: {'Excellent': 1, 'Very Good': 3, 'Good': 1}

citation_statistics: {'highest': 171, 'lowest': 42, 'average': 90.8, 'total': 454}

highest_cited_title: Development and validation of an autonomous artificial intelligence agent for clinical decision-making in oncology

latest_paper_title: Development and validation of an autonomous artificial intelligence agent for clinical decision-making in oncology

Research Gap Analysis

Total Papers: 5

Research Areas

• General Artificial Intelligence

• Machine Learning

• Retrieval-Augmented Generation

Common Keywords

• artificial

• agent

• application

• development

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• responsibility

• rock

• sdlc

• situation

• software

• stage

• sustainability

• sustainability-related

• sustainable

• three

• tools

• transformative

• transformer

• tunnelling

• web-based

Future Work

• While expectations for AI’s transformative role in sustainable development are high, its full potential remains to be realized.

Research Area Distribution

Machine Learning: 2

General Artificial Intelligence: 2

Retrieval-Augmented Generation: 1

Keyword Frequency

artificial: 4

agent: 2

application: 2

development: 2

intelligence: 2

vision: 2

accuracy: 1

actionable: 1

address: 1

ai-based: 1

ai-driven: 1

applied: 1

appropriate: 1

assistant: 1
attributed: 1
autonomous: 1
belt: 1
clinical: 1
clustering: 1
concept: 1
conceptually: 1
conducted: 1
control: 1
cycle: 1
data-driven: 1
decision-making: 1
deep: 1
design: 1
dimension: 1
function: 1
goal-related: 1
governance: 1
high-speed: 1
human: 1
image: 1
interview: 1
knowledge: 1
learning: 1
life: 1
making: 1
management: 1
mass: 1
meaningful: 1
muck: 1
mucks: 1
multimodal: 1
oncology: 1
optimization: 1
personalized: 1
project: 1
real-time: 1

remain: 1

responsibility: 1

rock: 1

sdlc: 1

situation: 1

software: 1

stage: 1

sustainability: 1

sustainability-related: 1

sustainable: 1

three: 1

tools: 1

transformative: 1

transformer: 1

tunnelling: 1

web-based: 1

Research Trends

• artificial

• agent

• application

• development

• intelligence

• vision

• accuracy

• actionable

• address

• ai-based

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Gap Categories

datasets: []

models: ['Clinical decision-making in oncology is complex, requiring the integration of multimodal data and multidomain expertise. We developed and evaluated an autonomous clinical artificial intelligence (AI) agent leveraging GPT-4 with multimodal precision oncology tools to support personalized clinical decision-making. The system incorporates vision transformers for detecting microsatellite instability and KRAS and BRAF mutations from histopathology slides, MedSAM for radiological image segmentation and web-based search tools such as OncoKB, PubMed and Google. Evaluated on 20 realistic multimodal patient cases, the AI agent autonomously used appropriate tools with 87.5% accuracy, reached correct clinical conclusions in 91.0% of cases and accurately cited relevant oncology guidelines 75.5% of the time. Compared to GPT-4 alone, the integrated AI agent drastically improved decision-making accuracy from 30.3% to 87.2%. These findings demonstrate that integrating language models with precision oncology and search tools substantially enhances clinical accuracy, establishing a robust foundation for deploying AI-driven personalized oncology support systems. Ferber et al. present an autonomous artificial intelligence agent system for deployment of specialized medical oncology computational tools, validating their system across various clinical scenarios representative of typical patient care workflows.', 'These findings demonstrate that integrating language models with precision oncology and search tools substantially enhances clinical accuracy, establishing a robust foundation for deploying AI-driven personalized oncology support systems.', 'In this study we explore the incorporation of artificial intelligence (AI) governance to system development life cycle (SDLC) models. We conducted expert interviews among AI and SDLC professionals and analyzed the interview data using qualitative coding and clustering to extract AI governance concepts. Subsequently, we mapped these concepts onto three stages in the machine learning (ML) system development process: (1) design, (2) development, and (3) operation. We discovered 20 governance concepts, some of which are relevant to more than one of the three stages. Our analysis highlights AI governance as a complex process that involves multiple activities and stakeholders. As development projects are unique, the governance requirements and processes also vary. This study is a step towards understanding how AI governance is conceptually connected to ML systems' management processes through the project life cycle. CCS CONCEPTS • Software and its engineering $^{\\rightarrow}$ Software creation and management.', 'In this study we explore the incorporation of artificial intelligence (AI) governance to system development life cycle (SDLC) models.']

retrieval: ['Retrieval-Augmented Generation']

grounding: []

hallucination: []

knowledge_freshness: []

evaluation: ['Clinical decision-making in oncology is complex, requiring the integration of multimodal data and multidomain expertise. We developed and evaluated an autonomous clinical artificial intelligence (AI) agent leveraging GPT-4 with multimodal precision oncology tools to support personalized clinical decision-making. The system incorporates vision transformers for detecting microsatellite instability and KRAS and BRAF mutations from histopathology slides, MedSAM for radiological image segmentation and web-based search tools such as OncoKB, PubMed and Google. Evaluated on 20 realistic multimodal patient cases, the AI agent autonomously used appropriate tools with 87.5% accuracy, reached correct clinical conclusions in 91.0% of cases and accurately cited relevant oncology guidelines 75.5% of the time. Compared to GPT-4 alone, the integrated AI agent drastically improved decision-making accuracy from 30.3% to 87.2%. These findings demonstrate that integrating language models with precision oncology and search tools substantially enhances clinical accuracy, establishing a robust foundation for deploying AI-driven personalized oncology support systems. Ferber et al. present an autonomous artificial intelligence agent system for deployment of specialized medical oncology computational tools, validating their system across various clinical scenarios representative of typical patient care workflows.', 'Experimental Evaluation', 'We developed and evaluated an autonomous clinical artificial intelligence (AI) agent leveraging GPT-4 with multimodal precision oncology tools to support personalized clinical decision-making.']

applications: ['Clinical decision-making in oncology is complex, requiring the integration of multimodal data and multidomain expertise. We developed and evaluated an autonomous clinical artificial intelligence (AI) agent leveraging GPT-4 with multimodal precision oncology tools to support personalized clinical decision-making. The system incorporates vision transformers for detecting microsatellite instability and KRAS and BRAF mutations from histopathology slides, MedSAM for radiological image segmentation and web-based search tools such as OncoKB, PubMed and Google. Evaluated on 20 realistic multimodal patient cases, the AI agent autonomously used appropriate tools with 87.5% accuracy, reached correct clinical conclusions in 91.0% of cases and accurately cited relevant oncology guidelines 75.5% of the time. Compared to GPT-4 alone, the integrated AI agent drastically improved decision-making accuracy from 30.3% to 87.2%. 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Deep learning and supervised machine learning were the most prominently applied algorithms for forecasting and system optimization. However, we identify a critical gap: only a few studies combine advanced AI applications with deep sustainability expertise. Sustainability needs to strike a balance between contextualization and generalizability to provide tangible knowledge that will lead to responsible change. AI must play a central role in this process. While expectations for AI's transformative role in sustainable development are high, its full potential remains to be realized. The use of artificial intelligence (AI) in sustainability-related scholarly work, such as Sustainable Development Goal research, is growing. An analysis now finds that few studies actually use AI to address normative or transformative dimensions of sustainability science, limiting the potential of relevant AI applications.', 'However, we identify a critical gap: only a few studies combine advanced AI applications with deep sustainability expertise.', 'How can humans remain in control of artificial intelligence (AI)-based systems designed to perform tasks autonomously? Such systems are increasingly ubiquitous, creating benefits - but also undesirable situations where moral responsibility for their actions cannot be properly attributed to any particular person or group. 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Second, humans and AI agents within the system should have appropriate and mutually compatible representations. Third, responsibility attributed to a human should be commensurate with that human's ability and authority to', 'Abstract Intelligent tunnelling has become an important direction for the development of TBM technology recently. As a result of the interaction between rock mass and TBM cutterhead, mucks are very important for predicting rock mass conditions and evaluating rock breaking efficiency. A real-time muck analysis system for assistant intelligence TBM tunnelling is proposed in this paper. Machine vision was applied to take the muck images continuously in the high-speed conveyor belt. The image segmentation and feature extraction of the mucks are conducted by using a deep learning algorithm. The proposed system also measured the mass and volume flow of the muck by installing a belt scale and a scanner to monitor the stability of the rock mass on the tunnel face. After the system was completed, it was installed on an indoor simulation experimental platform. A series of experiments were conducted to verify the design functions and measurement accuracy. Additionally, the system was applied to a TBM tunnelling project. The application results showed that the proposed system reached its design requirements and functions, and can provide muck data support for further assistant intelligent TBM tunnelling.']

coordination: []

security: []

scalability: ['Abstract Intelligent tunnelling has become an important direction for the development of TBM technology recently. As a result of the interaction between rock mass and TBM cutterhead, mucks are very important for predicting rock mass conditions and evaluating rock breaking efficiency. A real-time muck analysis system for assistant intelligence TBM tunnelling is proposed in this paper. Machine vision was applied to take the muck images continuously in the high-speed conveyor belt. The image segmentation and feature extraction of the mucks are conducted by using a deep learning algorithm. The proposed system also measured the mass and volume flow of the muck by installing a belt scale and a scanner to monitor the stability of the rock mass on the tunnel face. After the system was completed, it was installed on an indoor simulation experimental platform. A series of experiments were conducted to verify the design functions and measurement accuracy. Additionally, the system was applied to a TBM tunnelling project. The application results showed that the proposed system reached its design requirements and functions, and can provide muck data support for further assistant intelligent TBM tunnelling.']

explainability: []

reasoning: []

memory: []

planning: []

general: ['While expectations for AI’s transformative role in sustainable development are high, its full potential remains to be realized.', 'How can humans remain in control of artificial intelligence (AI)-based systems designed to perform tasks autonomously?']

Research Gaps

• Retrieval quality and the ability to consistently retrieve relevant information for complex queries remain important research challenges.

• More robust and standardized evaluation and benchmarking methods are required.

• Current models and architectures still require improvements in reliability, adaptability, and generalization.

• Scalability, computational cost, and efficient resource usage remain challenges for practical deployment.

• Further validation in real-world applications and deployment environments is required.

• Experimental Evaluation

• Retrieval-Augmented Generation

• While expectations for AI’s transformative role in sustainable development are high, its full potential remains to be realized.

• How can humans remain in control of artificial intelligence (AI)-based systems designed to perform tasks autonomously?

• The concept of meaningful human control has been proposed to address responsibility gaps and mitigate them by establishing conditions that enable a proper attribution of responsibility for humans; however, clear requirements for researchers, designers, and engineers are yet inexistent, making the development of AI-based systems that remain under meaningful human control challenging.

Emerging Topics

• artificial

• agent

• application

• development

• intelligence

• vision

Recommendations

• Explore more reliable autonomous and multi-agent systems with improved coordination and task execution.

• Improve retrieval quality and document selection for complex research queries.

• Evaluate proposed approaches using standardized benchmarks and reproducible evaluation protocols.

• Compare stronger model architectures and established baselines to identify improvements in reliability and generalization.

• Optimize computational cost and resource usage to improve scalability for practical deployment.

• Validate the proposed approaches in realistic real-world environments to assess scalability and practical usability.

• Compare proposed approaches with strong machine learning baselines using consistent evaluation metrics.

• Prioritize future research directions identified across the analyzed papers and validate them through controlled experiments.

Recommendation Validation

• {'recommendation': 'Explore more reliable autonomous and multi-agent systems with improved coordination and task execution.', 'validation_status': 'Partially Supported', 'evidence_score': 1, 'evidence': {'research_gaps': False, 'future_work': False, 'emerging_topics': True, 'methodology': False}}

• {'recommendation': 'Improve retrieval quality and document selection for complex research queries.', 'validation_status': 'Supported', 'evidence_score': 2, 'evidence': {'research_gaps': True, 'future_work': False, 'emerging_topics': False, 'methodology': True}}

• {'recommendation': 'Evaluate proposed approaches using standardized benchmarks and reproducible evaluation protocols.', 'validation_status': 'Supported', 'evidence_score': 2, 'evidence': {'research_gaps': True, 'future_work': False, 'emerging_topics': False, 'methodology': True}}

• {'recommendation': 'Compare stronger model architectures and established baselines to identify improvements in reliability and generalization.', 'validation_status': 'Supported', 'evidence_score': 2, 'evidence': {'research_gaps': True, 'future_work': False, 'emerging_topics': False, 'methodology': True}}

• {'recommendation': 'Optimize computational cost and resource usage to improve scalability for practical deployment.', 'validation_status': 'Partially Supported', 'evidence_score': 1, 'evidence': {'research_gaps': True, 'future_work': False, 'emerging_topics': False, 'methodology': False}}

• {'recommendation': 'Validate the proposed approaches in realistic real-world environments to assess scalability and practical usability.', 'validation_status': 'Partially Supported', 'evidence_score': 1, 'evidence': {'research_gaps': True, 'future_work': False, 'emerging_topics': False, 'methodology': False}}

• {'recommendation': 'Compare proposed approaches with strong machine learning baselines using consistent evaluation metrics.', 'validation_status': 'Supported', 'evidence_score': 2, 'evidence': {'research_gaps': True, 'future_work': False, 'emerging_topics': False, 'methodology': True}}

• {'recommendation': 'Prioritize future research directions identified across the analyzed papers and validate them through controlled experiments.', 'validation_status': 'Partially Supported', 'evidence_score': 1, 'evidence': {'research_gaps': False, 'future_work': False, 'emerging_topics': False, 'methodology': True}}

Summary

dominant_area: Machine Learning

top_trend: artificial

future_work_items: 1

research_gap_count: 10

validated_recommendation_count: 8

Experiment Plan

Research Areas

• General Artificial Intelligence

• Machine Learning

• Retrieval-Augmented Generation

Recommended Datasets

• BEIR

• HotpotQA

• Hugging Face Datasets

• Kaggle Datasets

• MS MARCO

• Natural Questions

• OpenML

• TriviaQA

• UCI Machine Learning Repository

Baseline Models

• BM25 Retrieval

• Decision Tree

• Dense Retrieval RAG

• Hybrid Search RAG

• Logistic Regression

• Naive RAG

• Random Forest

• Transformer

• Vanilla RAG

• XGBoost

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- Retrieval Precision
- Retrieval Recall
- Context Relevance
- Answer Relevance
- Faithfulness
- Groundedness
- ROC-AUC

Hardware Requirements

cpu: Intel Core i7 / AMD Ryzen 7

ram: 32 GB

gpu: NVIDIA RTX 3060 or better

storage: 200 GB SSD

Validation Strategy

- Train-Test Split
- K-Fold Cross Validation
- Hyperparameter Tuning
- Statistical Significance Testing
- Retrieval Evaluation
- Answer Quality Evaluation

Experimental Workflow

- Collect Knowledge Documents
- Preprocess and Chunk Documents
- Build Document Index
- Configure Retrieval System
- Retrieve Relevant Context
- Generate Grounded Response
- Evaluate Retrieval Quality
- Evaluate Answer Quality
- Compare Baseline and Proposed RAG
- Analyze Errors

• Draw Conclusions

Gap Based Recommendations

• Compare BM25, dense retrieval, and hybrid retrieval methods.

• Use standardized benchmarks and multiple evaluation metrics.

• Measure computational cost, latency, resource consumption, and scalability.

• Validate the system in realistic real-world deployment scenarios.

Report Summary

Dominant Research Area: Machine Learning

Top Research Trend: artificial

Highest Cited Paper: Development and validation of an autonomous artificial intelligence agent for clinical decision-making in oncology

Latest Paper: Development and validation of an autonomous artificial intelligence agent for clinical decision-making in oncology

Recommended Datasets

• BEIR

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Evaluation Metrics

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- Retrieval Precision
- Retrieval Recall
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- Answer Relevance
- Faithfulness
- Groundedness
- ROC-AUC

Research Recommendations

- Explore more reliable autonomous and multi-agent systems with improved coordination and task execution.
- Improve retrieval quality and document selection for complex research queries.
- Evaluate proposed approaches using standardized benchmarks and reproducible evaluation protocols.
- Compare stronger model architectures and established baselines to identify improvements in reliability and generalization.
- Optimize computational cost and resource usage to improve scalability for practical deployment.
- Validate the proposed approaches in realistic real-world environments to assess scalability and practical usability.
- Compare proposed approaches with strong machine learning baselines using consistent evaluation metrics.
- Prioritize future research directions identified across the analyzed papers and validate them through controlled experiments.

Research Gap Count: 10

Future Work Items: 1

Conclusion

The report has been generated using a personalized multi-agent pipeline. The explanation style was adapted according to the Intermediate profile to improve readability and usefulness.